# Explainability in Entity-Aware Financial Sentiment Classifiers

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## Task Description

Explain classification decisions made by an entity-aware sentiment analysis model using counterfactual explanations and feature attribution.

## Research Question

- How can we best use attribution-guidance to generate valid, plausible, and diverse counterfactual explanations in this context?
  - **Validity** A counterfactual is valid if it flips the class label of only the target entity while minimizing edits to the sample.
  - **Plausibility** The CFE should present a scenario that is within the scope of possibility. A counterfactual is considered plausible if it exhibits a high degree of similarity to existing samples in the dataset. This is important to ensure the explanations provide recourse (actionable information) to the end-user.
  - **Diversity**: A set of counterfactuals are considered diverse if the token-distance between pairs of counterfactuals is high. This helps give insight into different parts of the input sample that could be modified to change the outcome.

## Definitions

- Counterfactual Explanation (CFE): A method of explaining an AI's decision using a counter example. Stated like "If you had said it in this way I would classify differently"
- XAI: Explainable Artificial Intelligence
- LIME: Locally Interpretable Model Explanations
- Feature Attribution: The parts of the input that had the biggest effect on the prediction
- Entity-Aware: A model trained to predict the class of an entity (person/place/thing) P(class | text, entity)
- Sentiment Analysis: Using algorithms to assign positive, negative, or netural labels to text

## Challenges / Motivation

- Financial text contains complex, domain-specific terminology; necessitating a deeper understanding of models that deal with it.
- Financial text often refers to multiple entities, causing confusion among sentence-level classification models.
- Lanuage models used in high-stakes environments must be explainable in order to ensure trust and transparency.

## Example

**Counterfactual generation (first-shot, using Gemma3N: 4B) with and without feature attribution:**

**Original text:** "Microsoft plunges, Meta rallies as investors demand AI payoffs" - Reuters
**Entity:** Microsoft **Class:** Negative

**Feature Attribution Table (Most important words)**

| | | |
|---|---|---|
| Plunges (0.89) | rallies (-0.11) | demand (0.06) |
| Microsoft (0.05) | payoffs (0.04) | AI (0.03) |
| investors (0.02) | as (0.02) | Meta (-0.01) |

**Feature Attribution:** "Microsoft surges, Meta rallies as investors demand AI payoffs"
**Baseline (No Attribution):** "Microsoft surges, Meta dips as investors are satisfied with AI progress."

| Criteria | Valid | Plausible | Minimally changed |
|---|---|---|---|
| With Feature Attribution | Yes | Yes | Yes |
| Baseline (No Attribution) | No | No | No |

**Interpretation:** Using feature attribution, we can generate counterfactual explanations for these models that are more useful and effective.

## Visualization